

From Cars to Clinics

Service-Oriented Middleware for Networked Medical Devices

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A patient at home. A clinician in the hospital.

One question drives the next 20 minutes

How does the dashboard actually talk to the sensor — safely, interoperably, in real time?

Prerequisites I assume

Semester	Course	What I will reuse
3	Kommunikationsnetze	OSI, TCP/UDP, HTTP, sockets
3	Multimodale Signalverarbeitung	Sampling, sensor data streams
4	Verteilte Systeme	SOA, multi-tier, security
4	Softwaretechnik	IEC 62304 software lifecycle
4	Standards und Prozesse	Medical workflow vs. informatics

Requirements

What a networked medical device must offer

1. **Interoperability** — multi-vendor plug-and-play in the OR / at home
2. **Real time** — alarms, closed-loop control, streaming vitals
3. **Safety** — IEC 62304, ISO 14971, IEC 60601-1
4. **Security** — IEC 62443, MDR, threat modelling
5. **Traceability** — who did what, when, from which device
6. **Scalability** — device → room → hospital → network

A single scenario forces all six

- Wearable sends *TidalVolume*, *HeartRate*, *StepCadence*.
- Dashboard subscribes, renders, raises an alert if a threshold is crossed.
- Multiple vendors. Multiple networks. Multiple regulatory jurisdictions.

These are the non-functional requirements. They drive the architecture.

IEEE 11073 — a family, not a single standard

Part	Scope
11073-10207	Domain Information & Service Model (BICEPS)
11073-20701	Service-Oriented Device Connectivity (SDC)
11073-20702	Medical DPWS binding (transport)
11073-10101(nu)	Nomenclature codes for vitals, alarms, events

BICEPS = Basic Integrated Clinical Environment Protocol Specification.

The three roles in SDC

- **Provider** — a medical device that offers services (a ventilator).
- **Consumer** — a system that consumes them (a monitor, a dashboard).
- **SDC network** — discoverable, self-describing, secured

Automotive lens

Same problem, different domain

In eight years at Bosch I built middleware for a car with 100+ ECUs. The requirements list I showed you earlier?

Word for word the same, minus one column.

Side-by-side

Concern	Automotive	Medical
Service-oriented comms	SOME/IP (AUTOSAR AP)	SDC (IEEE 11073-20701)
Real-time streaming	DDS, iceoryx (zero-copy)	DDS, SDC waveform states
Safety lifecycle	ISO 26262 (ASIL A-D)	IEC 62304 (Class A/B/C) + ISO 14959
Cybersecurity	ISO/SAE 21434	IEC 62443 + MDR
Process maturity	Automotive SPICE	ISO 13485 + IEC 62304
Risk acceptance	Residual risk only	Residual risk vs. clinical benefit

Where SDC fits

One wearable, one dashboard — but there is more

The demo you have been watching sits at the bottom of a much taller stack that a networked medical system spans:

Physician office / EHR / national telematics platform

FHIR / TI-Messenger / KIM (system-to-system)

Hospital gateway

SDC / IEEE 11073 (device-to-device)

Bedside monitor, ventilator, imaging, robotics

What I would bring to CaMed

Three concrete directions — each with an existing anchor at Reutlingen:

1. **SDC as the substrate for the connected OR** — image-guided procedures, real-time device orchestration, 5G-connected surgical scenarios.
2. **Digital-twin-based V&V of networked device configurations** — carrying over the Bosch/ETH work on automotive digital twins.
3. **Uncertainty quantification for sensor-fusion pipelines** — from wearables to closed-loop clinical decisions (my PhD, applied to medicine).

Teaching stance

- Project-based, end-to-end: requirements → architecture → code → V&V → regulatory dossier.
- Everything as code, everything reproducible, everything reviewable.

BICEPS metric types in detail

- `NumericMetricState` — single scalar with unit + timestamp
- `StringMetricState` — textual value
- `EnumStringMetricState` — enumerated value
- `RealTimeSampleArrayMetricState` — waveform, sampled data
- `DistributionSampleArrayMetricState` — statistical distributions

IEC 62304 Class C artifact list

- Software development plan
- Software requirements specification
- Software architecture
- Detailed design
- Unit implementation + verification
- Integration + integration testing